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Multiplex analysis of ovarian cancer patients using glycan microarray

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ABSTRACT

Investigation of tumor-associated glycan antigens (TAGs) could be helpful for the development of sensitive cancer diagnostics and novel therapies. Glycan microarrays are effective methods for analyzing glycans and anti-glycan antibodies, which are immobilized arrays of oligo- or poly-saccharides on different substrates, making them a promising class of oncological biomarkers. Blood serum samples from patients (n = 203) with ovarian cancer (OvaCan) and healthy volunteers were analyzed using a glycan microarray containing 63 immobilized glycans to determine changes in anti-glycan IgG and IgM antibody profiles in OvaCan. Levels of anti-glycan IgG and IgM antibodies in OvaCan statistically differed from levels in healthy donors: the most prominent statistically significant difference for anti-glycan IgG antibodies was found for 6-*O*-su-Le^c (AUC = 0.657, Se = 48.0 %, and Sp = 73.3 %). The AUC values for certain glycans investigated in diagnosing OvaCan indicated a fingerprint consisting of IgM antibodies to specific glycans, and the most specific anti-glycan IgM antibodies were Le^y (AUC = 0.625, Se = 98.0 % and Sp = 45.0 %). The potential of these serological biomarkers to distinguish between OvaCan and other malignancies is still an unresolved issue that requires more large-scale studies to confirm and validate the use of these biomarkers in the diagnosis of different types of cancer.

1. Introduction

Glycan structures can influence a wide range of natural processes, including protein-protein interactions, migration, folding of proteins, immunological recognition, cell adhesion, and inter-cellular signaling [1,2]. The repertoire of circulating anti-glycan antibodies in healthy individuals is relatively unique and stable, allowing for the monitoring of changes in protecting antibodies in diseases [3]. The profile of anti-glycan antibodies is influenced by environmental factors or previous diseases and includes antibodies against foreign antigens as well as auto-antibodies, including antibodies that bind to tumor-associated glycans (TAGs) [4]. These TAGs are expressed on the surface of cancer cells and can serve as potential targets for cancer immunotherapy [5]. The presence of antibodies that specifically bind to TAGs can be a valuable diagnostic and prognostic tool for detecting and monitoring cancer progression [6]. TAGs are formed by several aberrant glycosylation processes and are appropriate for specific tumor-targeted treatment [7].

The vast majority of traditional immunohistochemical methods allow for the investigation of only one marker at a time in a clinical sample and require significant amounts of material [8,9]. Enzyme-linked immunosorbent assay (ELISA), immunohistochemistry (IHC) and flow cytometer assays [10] are most common diagnostic tools, but their clinical applications are limited to autoimmune and infectious diseases. Glycan microarrays have provided several advantages over conventional immunoassay methods for analysis of carbohydrate-protein interactions, including: (1) The ability to screen in parallel several thousand binding events on a single slide; (2) A minimal amount of precious samples and expensive or scarce carbohydrates required for one experiment; (3) The multivalent display of glycans on the chip that mimics the presentation of carbohydrates in nature; and (4) Ensured comprehensive and reproducible results [11]. Mass spectrometry-based methods (MS), another traditional immunohistochemical method, provide high sensitivity and specificity for identifying and quantifying complex tumor-associated glycans [12]. This emerging technology in biomarker research overcomes limitations like loss of crucial histological information, localized differences, and inability to correlate results with traditional histology [12]. However, MS can be

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time-consuming and requires specialized training to interpret the data accurately [12]. Some of these challenges have been overcome by the development of glycan microarrays as a high-throughput technology for studying carbohydrate interactions, which has greatly contributed to our understanding of the biological roles of carbohydrates or glycoconjugates and their interactions with a variety of macromolecules [11].

Glycan microarrays provide a high-throughput analysis method for detecting OvaCan in tiny sample sizes, enabling a comprehensive assessment of OvaCan patients [13], which leads to the discovery of specific glycan biomarkers that are more sensitive and specific for early detection or monitoring of the disease. However, glycan microarray requires specialized equipment and expertise, which could be a limitation for some research labs.

As a result, glycan microarrays have evolved into an effective tool for researching carbohydrate-protein interactions. Herein, we have developed a discovery platform for selective detection of cancer-associated antibodies directed in patients with OvaCan to aberrant 63 different glycans, including O-glycans (Thomsen-Friedenreich (TF) and its precursor (Tn) and sialyl-Tn), globo-series glycans (Globo-H), Lewis's antigens (Le^a, Le^b, Le^X, and Le^Y), polysialic acid, and others, by using glycan microarray.

2. Materials and methods

2.1. Subjects

Blood serum samples from patients (n = 203) with OvaCan and healthy volunteers were examined to determine the concentrations of IgM and IgG antibodies to glycans: 154 OvaCan patients at various stages of the tumor and 49 donors as a control group, divided into two groups: 29 patients with benign tumors and 20 healthy volunteers (Fig. 1; Table 1).

Blood serum samples were collected in the clinics of the P.A. Hertsen Moscow Research Oncological Institute—National Medical Research Radiological Center of the Ministry of Health of the Russian Federation and the N.N. Blokhin National Medical Research Center of Oncology, after informed consent was obtained from each patient. Samples were collected using standard procedures from these patients before starting any treatment. Blood serum samples were drawn into sterile Vacutainer tubes and allowed to clot for 30 min at ambient temperature. The serum layer was removed, centrifuged, aliquoted, and stored at −80 °C. The data from the patients' sera were analyzed anonymously.

The control group in the study was selected through specific inclusion criteria that excluded individuals with malignancies or significant

Table 1
The characteristics of groups under study.

Group	n (Females)	Median age (range)
Healthy donors	20	42 (22–75)
Benign tumors patients	29	48 (21–75)
OvaCan patients:	154	58 (23–82)
Stage 1	71	55 (23–77)
Stage 2	15	63 (43–82)
Stage 3	62	54 (36–72)
Stage 4	6	61 (44–77)

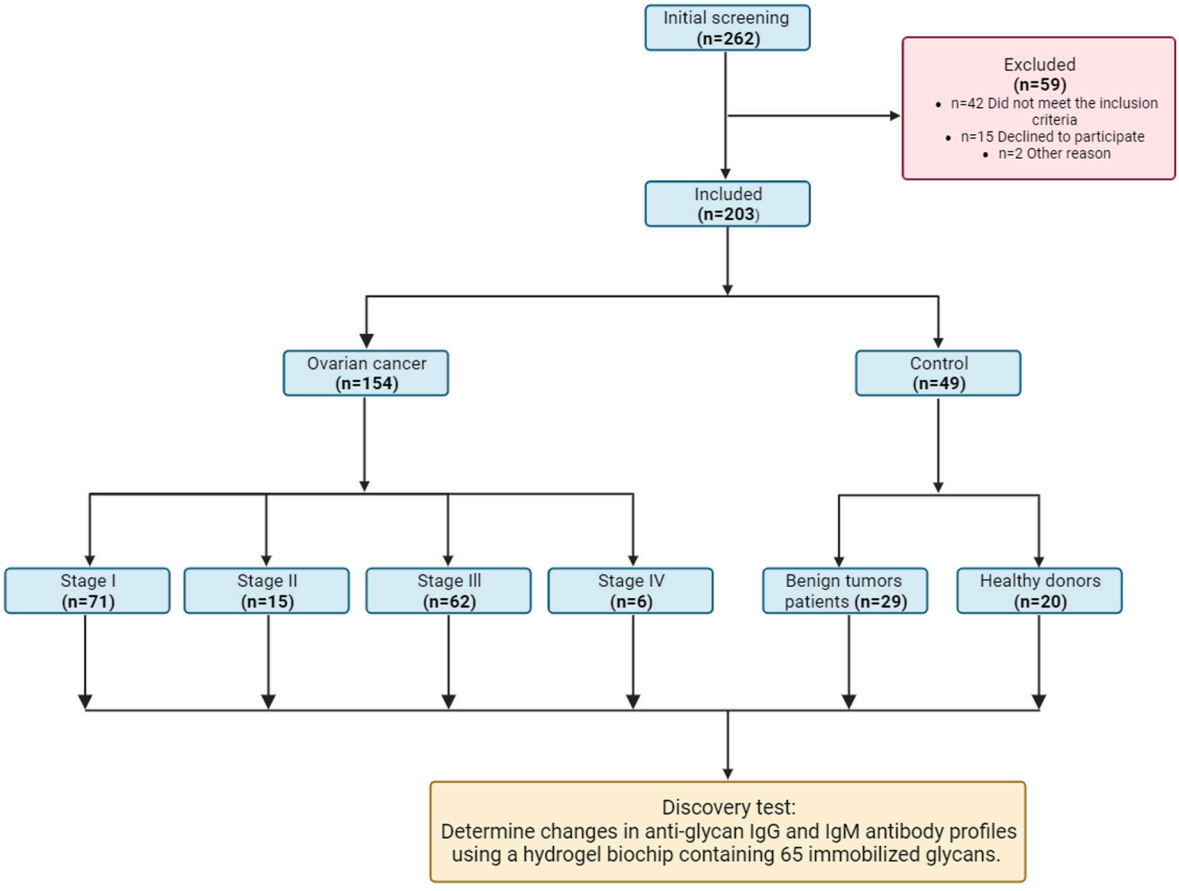


Fig. 1. STARD flow diagram of the study. Patient recruitment was performed on occasional days when the researchers were present in the hospital. The hospital nursing and medical staff informed the researchers about patients with diagnosed ovarian cancer (OvaCan) and probably fulfilling the inclusion criteria. A total of 203 blood serum samples from patients with OvaCan and healthy volunteers were screened in the participating hospitals by the researchers. At first, 59 patients were excluded due to failure to meet the inclusion criteria, refusal to participate in the study, or other reasons. The other reasons included equipment malfunctions.

health issues. Healthy volunteers were selected from the same population as the patients with ovarian cancer, minimizing selection bias, and efforts were made to match the control group in terms of age and sex. Health screenings confirmed that volunteers had no underlying conditions affecting antibody levels. A sufficient sample size was established for reliable statistical comparisons, and data were analyzed anonymously to prevent bias.

2.2. Analysis of blood sera on glycan microarray

The 3D glycan array was created using a technology developed at the Engelhardt Institute of Molecular Biology of the Russian Academy of Sciences and was designed to quantify antibody levels against 63 glycans (Supplementary Table 1) in a single experiment. The approach has a wide dynamic range and can quantify both low and high antibody titers [14,15]. Glycan microarrays were prepared and ensured their quality control in accordance with previous study [14,15]. Briefly, the proteins and oligosaccharides intended for immobilization were modified with the appropriate spacers and added to the polymerization mixture along with gel-forming methacrylamide-based monomers. Using a Qarray pin robot (Genetix, United Kingdom), microdroplets (0.1 nL) of the combination were applied to glass slides that had been activated with Bind-Silane. Arrays were exposed to UV light with a maximum at 350 nm from a Sylvania GTE lamp (F15T8/350Bl, United Kingdom) that was 8 cm distant from the arrays and at 20 °C for 50 min in order to allow gel cells to polymerize. The arrays were rinsed with distilled water after being polymerized and then washed with PBST (Phosphate-buffered saline (PBS), (pH = 7.4) + 0.1 % Tween-20) for 40 min. The final 3D gel elements have a diameter of 120 μm. Using TestChip and QualityControl software (Engelhardt Institute of Molecular Biology of the Russian Academy of Sciences), the appearance of the arrays in transmitted light was used to evaluate their quality. Approved arrays were placed within 120-μL polypropylene incubation chambers and kept between 2 and 8 °C. Glycans were immobilized with spacers at a concentration of 3.3 mM. Sera were diluted 1:50 with Phosphate-buffered saline (PBS), 0.14 % polyvinyl alcohol (PVA), and 0.14 % polyvinylpyrrolidone (PVP) to guarantee that the bulk of antibodies are detectable while only a few components reach saturation levels.

2.3. Antibody conjugation with fluorescent dyes

Cyanine N-hydroxysuccinimide ester (Cy5 or Cy3; Healthcare, United Kingdom) (2 μL, 2 mg/mL in a dimethylformamide solution) was added to 75 μL of an anti-IgG or anti-IgM solution (1 mg/mL in 0.01 M bicarbonate buffer, pH 9.5), which were purchased from Imtek LLC (Moscow, Russia). The reaction was carried out in the dark at 22 °C for 1 h with continuous agitation (550 rpm). Excess dye was removed by gel filtration on a Micro Bio-Spin column filled with Sephadex G-25 and equilibrated with PBS. A staining protocol was selected so that the protein–dye molar ratio was 1:3. The working concentrations of the resulting anti-IgG-Cy3 and anti-IgM-Cy5 secondary antibodies in the mixture applied to glycan arrays were 0.005 mg/mL.

2.4. Immunoassays on glycan arrays

Serum samples from OvaCan patients, benign tumor patients, and healthy donors were diluted with dilution buffer (PBS, 0.14 % PVA, and 0.14 % PVP) and incubated with glycan arrays at 37 °C for 2 h [16,17]. The arrays were washed with PBST for 30 min, rinsed with water, dried in an air flow, and incubated with 120 μL of a secondary antibody mixture at 37 °C for 1 h. After final washing (PBST, 30 min), the arrays were rinsed with water and dried in an air flow.

Glycan arrays were examined using a Genepix 4200 scanner (Molecular Devices, United States) at 10 μm resolution. Genepix Pro 6.0 software (Molecular Devices) was used to calculate the fluorescent signal intensity for each cell using fluorescence intensities at λ = 635 nm (column F635 median in Genepix Pro 6.0) for anti-glycan IgG antibodies and λ = 532 nm (column F532 median in Genepix Pro 6.0) for anti-glycan IgM antibodies.

2.5. Statistical analysis and data presentation

In evidence-based medicine, the study of Receiver Operating Characteristic curve analysis (ROC curves) [18] is frequently used to assess a test system's prognostic capacity. The effectiveness of various diagnostic techniques in identifying diseases may be compared using this technique. Plotting true-positive rates versus false-positive rates discovered throughout the course of the investigation allows for the creation of ROC curves [19,20].

Using multiparameter log-regression analysis and ROC curves, a comparative analysis of the diagnostic importance of various components in multiplex measurements on glycan microarrays was carried out. The diagnostic effectiveness of the associated factor is inversely correlated with the area under the ROC curve (AUC). When AUC = 0.9–1.0, the diagnostic procedure is deemed excellent; AUC = 0.8–0.9, very good; AUC = 0.7–0.8, good; AUC = 0.6–0.7, satisfactory; and when AUC = 0.5–0.6, unsatisfactory.

For quantitative assessment of the efficacy of diagnostics, two cohorts of patients are selected: patients with OvaCan and healthy donors. The basic principle of the ROC analysis is the comparison between the sensitivity (Se) and specificity (Sp) of the model, calculated from the number of false-positive and false-negative diagnoses as we described previously. The calculations of Se and Sp operate not with absolute values, but with relative rates expressed as percentages. The true-positive rates (TPR) were determined using the following formula:

$$TPR\,\% = \frac{TP}{TP + FN}\,X\,100$$

whereas false-positive rates (FPR) were determined using the following formula:

$$FPR\,\% = \frac{FP}{FP + TN}\,X\,100$$

Where TP (true positives) are correct positive diagnoses; TN (true negatives) are correct negative diagnoses; FN (false negatives) are errors of the first type; FP (false positives) are errors of the second type.

The following formulas are used to get Se and Sp from the binary model using ROC analysis.

The Se of the model is the proportion of TP results:

$$Se = TPR\,\% = \frac{TP}{TP + FN}\,X\,100$$

The Sp of the model is the proportion of TN results:

$$Sp = 100 - FPR\,\% = \frac{FP}{FP + TN}\,X\,100$$

When the number of FN and FP outcomes approaches zero, Se and Sp in the binary model of diagnostics approach 100 % (or AUC approaches unity).

The results were statistically processed using the program MedCalc Version 11.4.2.0 (Ostend, Belgium), OvaCan = 1, benign tumor patients (b-Can) = 0, and healthy donors (H) = 0.

Data analysis, including calculation of mean, median, standard deviation and coefficient of variation was performed using the open-source statistical programming language R (http://CRAN.R-project.org/, version 2.8.1). A one-way analysis of variance (ANOVA) test was also used to evaluate the statistical significance between different groups under various circumstances using Tukey's multiple comparison test. The Tukey's test confirmed that there were statistically significant differences between the groups for some aberrant glycans. This is more proof that the independent and dependent variables are related in a

meaningful way. In order to ensure the accuracy of the results, each sample in the study was assayed twice, with three separate replicates taken for every condition that was tested. $p < 0.05$ was considered statistically significant.

3. Results

Based on specific binding between antibody-antigen pairs, immunoassay analysis is commonly performed in two types: sandwich-type immunoassay and competitive immunoassay [21]. Sandwich-type immunoassay offers high sensitivity and specificity due to its two binding sites on the antigen, making it suitable for detecting low quantities of antigens in complex samples and commonly used to quantification of large molecules [22]; nevertheless, there is a risk of cross reactivity [23]. On the other hands, competitive immunoassay is less sensitive to experimental errors as it only requires one binding site on the antigen, making it suitable to quantification free and/or small molecules [24]. Nevertheless, competitive immunoassay is quicker, more flexible and has good reproducibility [21].

In our study, we used a sandwich variant of immunofluorescent analysis to determine the anti-glycan antibody profiles in a blood serum sample on a glycan microarray. This method allows for simultaneous detection and quantification of multiple antigen-antibody interactions on a single chip, providing valuable information on glycan profiles in blood serum samples.

To implement the possibility of simultaneous detection of anti-glycan antibodies of two classes (IgG and IgM), the corresponding secondary antibodies were conjugated with Cy3 and Cy5 fluorescent dyes with different emission spectra. This approach made it possible to simultaneously obtain two sets of data on IgG and IgM antibodies specific to all 63 glycans during one analysis on one glycan microarray.

3.1. Evaluation of IgG levels in sera from patients with OvaCan and healthy donors

In general, we found significant differences between the serum levels of anti-glycan IgG antibodies (6-*O*-su-Le^c and P_1) in the control and experimental cohorts. The level of serum-derived naturally circulating anti-glycan IgG antibodies to immobilized sulfated trisaccharide (6-*O*-su-Le^c and P_1) has been increased by 2.0-fold ($p = 0.006$) and 2.2-fold ($p = 0.01$), respectively, in patients with OvaCan compared with the control group.

The differences in the serum level of anti-glycan IgG antibodies to other glycans (GlcNAc3′Le^c, 6-Su-3′NeuGcLe^c, GalNAcα4′LN, 6-Su-3′SiaLe^c, 6-SiaTF, 6-SiaβTF, 3,6-SiaTn, 3-LacNAc-Tn, Le^y, SiaLe^a, Sia-6-H, Zumozan A, Le^{c}3Le^c, and Le^c-α-6LN) were found between two cohorts when the differences were not statistically significant ($p > 0.05$). Regarding the other studied anti-glycan IgG antibodies, there were no differences in the level of IgG between the two cohorts.

The values of AUC for certain glycans studied in detecting OvaCan revealed a signature consisting of IgG antibodies to glycans. The most effective glycan to determine the OvaCan by anti-glycan IgG antibody was statistically significant with 6-*O*-su-Le^c (AUC = 0.657; Se = 48.0 % and Sp = 73.3 %).

No statistically significant differences in the serum level of anti-glycan IgG antibodies were seen between the healthy volunteers and benign tumor patients ($p > 0.05$) for all glycans.

3.2. Evaluation of IgM levels in sera from patients with OvaCan and healthy donors

To provide an overall view of the anti-glycan IgM antibody repertoires, the levels of anti-glycan IgM antibody signals were measured. We found a significant differences between the serum levels of anti-glycan IgM antibodies (TF, TF-long, Tαα, SiaTn, 3′O-su-Le^c, 3′O-su-TF, core 6, 6′O-su-Le^c, Neu5Gc-Tn, β-SiaTn, P_1, 3-LacNAc-Tn, 3′-*O*-su-Le^a, core 2, 3′-sialyl-TF, 6-Su-3′SiaLe^c, Le^y, SiaLe^a, Zumozan, Le^a, Le^b, Le^x, aminoglucitol, β-6-'SLN, LNβ6′LN, GN4′LN, and GN6′LN) in OvaCan patients and controls. The level of serum-derived naturally circulating anti-glycan IgM antibodies to immobilized fucosylated glycans (Le^a, Le^b, Le^x, Le^Y, SiaLe^a, and 3′-*O*-su-Le^a) has increased significantly by 2.4-fold ($p = 0.001$), 2.6-fold ($p = 0.001$), 2.4-fold ($p = 0.003$), 2.3-fold ($p = 0.003$), 1.7-fold ($p = 0.017$), and 1.9-fold ($p = 0.001$), respectively, in patients with OvaCan when compared with control.

Interestingly, the level of anti-glycan IgM antibodies (6-Su-3′SiaLe^c) in serum from OvaCan patients has significantly fallen by 2.7-fold ($p < 0.001$) when compared with controls. Anti-glycan IgM antibodies significantly increased in OvaCan patients by 1.8-fold ($p = 0.002$), 1.7-fold ($p = 0.002$), 2.2-fold ($p = 0.009$), 2.2-fold ($p = 0.002$), 2.0-fold ($p = 0.012$), 1.4-fold ($p = 0.014$), and 2.0-fold ($p = 0.007$) for core 2, 3′-sialyl-TF, Zumozan, β-6′SLN, LNβ6′LN, GN4′LN, and GN6′LN, respectively. The variation in the serum level of IgM antibodies to other studied glycans has been seen between the two cohorts when the differences were not statistically significant ($p > 0.05$).

The values of AUC for certain glycans studied in detecting OvaCan revealed the signature consisting of the IgM antibodies to glycans. The most effective glycans to determine the OvaCan by anti-glycan IgM antibody were Le^y (AUC = 0.625, Se = 98.0 % and Sp = 45.0 %).

The differences in the serum level of anti-glycan IgM antibodies between the healthy volunteers and benign tumor patients were not statistically significant ($p > 0.05$) for any of the studied glycans.

3.3. Differences in anti-glycan IgG/IgM antibodies in relation to the stage of ovarian cancer

The effect of stages of cancer on anti-glycan antibody profiles was surprisingly different for IgG and IgM antibodies. The levels of IgG antibodies to 6-*O*-su-Le^c and P_1 in the cohorts of patients with OvaCan stages I and healthy donors differed significantly ($p = 0.032$ and 0.008, respectively). The sensitivity and specificity of these biomarkers were 56.2 and 69.1 %, and 71.0 and 68.4 %, respectively.

In contrast, there are significant differences in anti-glycan IgM antibody (3′-*O*-su Le^a, 3′-sialyl-TF, 6-Su-3′SiaLe^c, and β-6′SLN) levels in serum control cohorts and patients with various stages of OvaCan ($p < 0.001$).

Significant differences in serum concentrations of anti-glycan IgM antibodies (3′-*O*-su Lea, and 3′-sialyl-TF) were found between healthy and patients with different stages of OvaCan (stage I, II, and III ($p < 0.001$)) but not for patients with stage IV.

For anti-β-6′SLN IgM antibody, there were significant differences between healthy and patients with all stages of OvaCan. However, significant differences were found for anti-glycan IgM antibody (6-Su-3′SiaLec) with p value < 0.001 between healthy and patients with stages I, and III. No significant differences were found for the other studied glycans.

4. Discussion

Immobilization of molecular probes on glycan microarrays provides significant benefits over typical flat surfaces and is regarded as a potential approach for cancer diagnosis [25,26]. Measurement of tumor markers in blood serum is widely used for early diagnosis of primary tumors and metastatic regions, as well as for treatment monitoring [3]. An ideal biomarker should be characterized by high specificity and sensitivity to a certain type of tumor. However, the majority of tumor markers do not satisfy these requirements [27].

Invasive histopathological biopsy is used as the gold standard diagnostic tool for detecting OvaCan. Several biomarkers were studied to develop a non-invasive diagnostic tool, out of which HE4 and CA 125 remain the only clinically useful biomarkers [28]. However, these biomarkers have limitations in terms of sensitivity and specificity, leading to the need to more study in developing more accurate and reliable

diagnostic tools for OvaCan. Glycan microarrays allow researchers to evaluate hundreds of extra anti-glycan IgG and IgM antibodies in order to find new ones with improved sensitivity and specificity [29]. Moreover, glycan microarrays can be utilized to quickly uncover combinations of biomarkers with clinically beneficial properties, including cancer detection, monitoring, and prognosis. The capacity to promptly analyze many samples, as demonstrated here, was helpful for finding anti-glycan IgG and/or IgM antibodies that were greater in serum samples from different cancer cohorts relative to controls using glycan microarrays [30].

Due to the lack of signs and symptoms of OvaCan in the early stages and the absence of an effective biomarker for diagnosis, OvaCan has the highest mortality rate of all gynecological cancers. For that, a microarray based on anti-glycan IgG and IgM antibodies was evaluated for its potential for diagnosing OvaCan [31].

Our glycan microarray was used to evaluate serum IgG and IgM from samples of 154 OvaCan patients and 49 volunteers. The array had 63 components, including a wide range of O-linked glycans, N-linked glycans, glycolipid glycans, and neoglycoproteins. There were some natural glycoproteins in the array as well. A wide variety of anti-glycan antibody populations in serum may therefore be found using the array.

In comparison to any single parameter, the multiplex analysis based on a particular set of several parameters demonstrated to be noticeably more accurate at diagnosing OvaCan. We discovered that the diagnostic effectiveness of both specific probes and the collection of probes significantly increased by adding the amount of immunoglobulin IgG and IgM as a diagnostic criterion. In patients with OvaCan compared to controls, the amount of serum-derived naturally circulating anti-glycan IgG and IgM antibodies to specific glycans has increased dramatically by approximately 2-fold in the cohort with OvaCan.

Neither glycans (6-*O*-su-Le^c) for IgG, nor glycans (3′-*O*-su-Le^a, core 2, 3′-sialyl-TF, 6-Su-3′SiaLe^c, Le^y, SiaLe^a, Zumozan, Le^a, Le^b, Le^x, β-6-'SLN, LNβ6′LN, GN4′LN, and GN6′LN) for IgM used in this study have been previously reported to be specific to OvaCan. The discovery of anti-glycan IgG and/or IgM antibodies in serum as biomarkers for cancer diagnosis has been demonstrated in several investigations; among the 24 glycan antibodies previously investigated, P_1 was the best candidate ($p < 0.001$) as a biomarker for OvaCan diagnosis [32].

This finding was validated in our investigation, where the levels of anti-P_1 antibodies were significantly higher in the OvaCan cohort as compared to the healthy cohort, indicating that P_1 is a promising candidate for OvaCan detection. Additionally, Pochechueva et al. previously found SiaTn and 6-OSulfo-TF to be useful alternatives to CA125 for the detection of ovarian cancer [33].

The differences in IgG and IgM antibody profiles suggest that these antibodies may reflect different aspects of the immune response to ovarian cancer. IgG antibodies may be more indicative of early-stage disease, while IgM antibodies could provide a broader indication of tumor presence in late-stages. The study implies that a combination of both IgG and IgM antibodies could enhance diagnostic accuracy and provide insights into the progression of ovarian cancer.

However, in the present study, we discovered that the levels of SiaTn were consistently higher in the experimental cohort compared to the control cohort. Although these differences did not reach statistical significance, they suggest a potential trend that warrants further investigation. Therefore, SiaTn could still be considered a promising biomarker for ovarian cancer detection, despite not meeting the threshold for statistical significance in this study. However, no difference was observed in the levels of antibodies to Globo H in serum from OvaCan patients compared to healthy donors, while there were significant differences in the level of antibodies to Globo H in serum from breast cancer patients compared to controls [34]. This finding contributes to the body of evidence suggesting that anti-glycan antibodies may have some perspective in identifying particular forms of cancer while ignoring others. The potential of these serological biomarkers to distinguish between OvaCan and other malignancies is still an unresolved issue that requires more large-scale studies to confirm and validate the use of these biomarkers in the diagnosis of different types of cancer. In future studies, we are planning to validate this capacity. However, monitoring glycosylation changes either directly or by measuring the levels of various anti-glycan IgG and/or IgM antibodies involves many limitations and challenges, particularly when evaluating samples from patients in the early stages of cancer.

Certain limitations of our study should be highlighted. (1) Since tumors escape immunological monitoring and have molecular heterogeneity, the pool of anti-glycan IgG and/or IgM antibodies varies frequently among tumors of the same stage and histological type, which reduces diagnostic test accuracy and increases the ratio of false-positive results. (2) Due to the high variability of the level of anti-glycan antibodies between individuals that is associated with age, race, sex, and blood group, anti-glycan antibody levels were found to be much higher in certain individuals than in others. (3) Unfortunately, we were unable to follow the prognosis of survival or effectiveness of treatment in these individuals, and therefore further studies with a larger sample size and a longer follow-up period are needed to evaluate the potential of this biomarker as a prognostic tool for treatment. (4) According to Ref. [35], serum IgG and IgM antibodies may compete for the same glycan antigen. Therefore, anti-glycan IgM antibodies may bind faster than anti-glycan IgG antibodies, and the experimental settings may not allow for adequate equilibration time. More advanced studies are needed to evaluate these possibilities.

5. Conclusion

Glycan microarrays allow simultaneous quantitative immunoassay of 63 aberrant glycans in a single experiment for the detection of OvaCan. The serum levels of anti-glycan IgG antibodies against glycan (6-*O*-su-Le^c) and anti-glycan IgM antibodies against glycan (Le^y) have been demonstrated to significantly correlate with the diagnosis of OvaCan. This may encourage the use of glycan microarrays in the diagnosis of OvaCan and other tumors, making them a promising tool for future cancer biomarker discovery. The potential of these serological biomarkers to distinguish between OvaCan and other malignancies is still an unresolved issue that requires more large-scale studies to confirm and validate the use of these biomarkers in the diagnosis of different types of cancer.

CRediT authorship contribution statement

Vadim S. Pokrovsky: Writing – original draft, Resources, Investigation, Data curation. **Louay Abo Qoura:** Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Aleksei A. Tikhonov:** Investigation, Data curation, Conceptualization. **Alla Yu Rubina:** Writing – review & editing, Supervision, Methodology. **Nikolai E. Kushlinskii:** Writing – review & editing, Supervision.

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Declaration of competing interest

All authors have read the journal's policy on disclosure of potential conflicts of interest and have none to declare. Moreover, all authors have read the journal's authorship agreement and reviewed and gave their final approval of the submitted version.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ab.2025.115806.

Data availability

Data will be made available on request.

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